



Contents lists available at ScienceDirect

J. Vis. Commun. Image R.

journal homepage: www.elsevier.com/locate/jvcir

Research on image steganography analysis based on deep learning[☆]

Ying Zou^a, Ge Zhang^{b,c,*}, Leian Liu^a^a College of Information Science and Technology, Zhongkai University of Agriculture and Engineering, No. 501, Zhongkai Rd, Haizhu District, Guangzhou, China^b School of Computer and Information Engineering, Henan University, Kaifeng 475004, China^c Henan Key Laboratory of Big Data Analysis and Processing, Henan University, Kaifeng 475004, China

ARTICLE INFO

Article history:

Received 27 August 2018

Revised 26 February 2019

Accepted 26 February 2019

Available online 26 February 2019

Keywords:

Steganalysis

Steganography

Feature learning

Deep learning

Convolutional neural network

Transfer learning

Multitask learning

ABSTRACT

Although steganalysis has developed rapidly in recent years, it still faces many difficulties and challenges. Based on the theory of in-depth learning method and image-based general steganalysis, this paper makes a deep study of the hot and difficult problem of steganalysis feature expression, and tries to establish a new steganalysis paradigm from the idea of feature learning. The main contributions of this paper are as follows: 1. An innovative steganalysis paradigm based on in-depth learning is proposed. Based on the representative deep learning method CNN, the model is designed and adjusted according to the characteristics of steganalysis, which makes the proposed model more effective in capturing the statistical characteristics such as neighborhood correlation. 2. A steganalysis feature learning method based on global information constraints is proposed. Based on the previous research of steganalysis method based on CNN, this work focuses on the importance of global information in steganalysis feature expression. 3. A feature learning method for low embedding rate steganalysis is proposed. 4. A general steganalysis method for multi-class steganography is proposed. The ultimate goal of general steganalysis is to construct steganalysis detectors without distinguishing specific types of steganalysis algorithms.

© 2019 Published by Elsevier Inc.

1. Introduction

With the rapid development of information technology and the rapid popularization of the Internet, digital media has become an important carrier for military, commercial and other organizations as well as individuals to obtain and transmit information. But at the same time, because the digital communication in the Internet is vulnerable to the threat of eavesdropping, malicious interference and other activities, people pay more attention to the security issues such as privacy protection and data integrity in the process of information transmission than ever before. The traditional solution uses encryption technology to convert the information to ciphertext for transmission. But its shortcoming is that the encrypted ciphertext is usually disordered. It is easy for an attacker to notice the existence of secret communication, which stimulates the attacker's enthusiasm for decoding. At the same time, it may also lead to information being interfered or intercepted, resulting in the failure of information transmission. In the above context, a new concept of communication security has been gradually

accepted and recognized: communication security means not only that the content of information to be transmitted is secure, but also that the existence of the act of transmitting secret information is unknown. Therefore, steganography, which is characterized by "camouflage" in the transmission of information, has attracted more and more attention.

The basic principle of steganography is to hide the secret information which needs to be transmitted into the redundant information by using the insensitive redundant information of human perception system existing in common carriers, and to realize the transmission of secret information by means of carrier transmission. Because the process of information hiding into the carrier usually does not change the normal perception characteristics of the carrier, it is difficult for potential attackers to detect the existence of secret information, thus ensuring the information security covert transmission. At the same time, it can also combine encryption, scrambling, coding and other technologies, making it difficult to extract hidden information even if it is detected by third parties, thus further ensuring the security of information transmission. In order to explain steganography more vividly, we describe the "prisoner problem" as an example. Alice and Bob are prisoners in different cells in the same prison. They are under the care of Warden Eve. Alice and Bob are going to discuss plans for a joint jail-break. Specific plans need to be negotiated through the exchange

[☆] This article is part of the Special Issue on TIUSM.

* Corresponding author at: School of Computer and Information Engineering, Henan University, Kaifeng 475004, China.

E-mail addresses: zouying@zhku.edu.cn (Y. Zou), zhangge@henu.edu.cn (G. Zhang), lla@zhku.edu.cn (L. Liu).

of information. However, according to prison regulations, their communications need to be checked by the warden, so they cannot communicate in plaintext. In this case, Bob and Mice need to take more covert communication measures. One consideration is to use encryption technology, that is, to hide the content of information, but because the encrypted information is a mess of code, it is easy to arouse Eve's suspicion. Thus, in this case, a safer consideration is to hide information in everyday objects, and make the hidden objects look normal, which can reduce Warden Eve's vigilance and ensure the smooth transmission of information.

However, steganography is also a true double-edged sword, which provides people with reliable and secure means of Internet communication, at the same time, it may also provide convenience for organizations and individuals with malicious intentions or improper purposes [11–14]. In fact, in recent years, there have been reports about the use of steganography in espionage, terrorist attacks, crimes and other activities. In 2001, some mainstream media in the United States, such as CNN and US Today, reported the news of secret communications between Al Qaeda members using steganography. Reported that bin Laden gang will attack the target map, action instructions and other information hidden in pornography, sports chat and other websites. According to Die Zeit, an al-Qaida suspect was arrested in Berlin in May 2011 and police found him carrying a memory card. Later, after being cracked by experts in charge of computer criminal investigation in the German Federal Criminal Police Bureau, it was found that on the surface, only one pornographic video named “KickAss” appeared on the card, but in fact, 141 text documents were hidden in the video, including a large number of Al-Qaida action reports, future action plans and so on. In June 2010, the Federal Bureau of Investigation (FBI) successfully arrested 10 Russian agents in New Jersey, which caused a great diplomatic shock between the United States and Russia.

The FBI said that by eavesdropping on the conversations between the Murphy couple, who were all suspects, they found that Murphy had sent his wife Cynthia to South America to hand over something “invisible” to someone. According to the FBI analysis, these so-called “invisible things” are likely to be recorded by digital steganography. They also claimed to have found devices for digital steganography in the homes of three suspected Russian agents, and inferred that Russian agents used steganography to communicate information with relevant Russian intelligence agencies. According to a Reuters report in July 2015, a report released by security manufacturer FireEye shows that Russian hackers have successfully invaded the U.S. defense system and captured several computers of the U.S. Department of Defense using data from Twitter that appears to be ordinary photos. These seemingly ordinary Twitter images hide information and instructions to activate malicious programs that have been implanted into the target computer. With the help of the camouflage of Twitter pictures, the transmission of commands controlling malicious programs easily avoids the detection of most detection systems. In April 2014, mainstream media in China, such as CCTV and Xinhua News Agency, reported that the mobile Trojan Horse “immortal Trojan Horse II” infected millions of mobile phones, and carried out fishing attacks, remote control of users' mobile phone photos, stealing users' online silver and other activities that seriously endangered users' personal privacy and property security. It is reported that, unlike the normal Trojan horse direct command mode, the Trojan horse cunningly disguises malicious code instructions as a common picture through steganography, in order to avoid the “pursuit” of mobile phone security software [1–10].

It can be seen from above that illegal or malicious use of digital steganography has brought serious harm to national information security, business and personal privacy and property security [15–19]. In this case, how to effectively supervise the use of

steganography in real life, prevent or block the malicious or illegal use of steganography in real time has become an urgent need of military and security departments in various countries. Because of this, Steganalysis, as a countermeasure technology of steganography, came into being, and has attracted the attention of governments and scientific research institutions. Steganalysis is a technique to determine whether there is additional information hidden in the carrier or not, even to estimate the amount of information embedded in the carrier, and to obtain the content of the hidden information by analyzing the statistical characteristics of the carrier. It plays the role of Warden Eve in the model of “prisoner problem” mentioned above. The research of steganalysis technology is of great significance in preventing the leakage of confidential information, combating terrorism and criminal activities, and maintaining Internet security.

Based on the theory of in-depth learning method and image-based general steganalysis as the research object, this paper focuses on the hot and difficult problem of feature expression in steganalysis, and tries to establish a new framework of steganalysis from the new idea of feature learning. The specific research contents of this paper mainly include the following aspects:

1. A digital image steganalysis framework based on depth learning is proposed.

Aiming at the problems in steganalysis, especially in feature expression. Considering feature learning, a new steganalysis paradigm based on in-depth learning is proposed. Based on the representative deep learning method CNN, the proposed model is designed and adjusted according to the characteristics of steganalysis, which makes the proposed model more effective in capturing the statistical characteristics such as neighborhood correlation Related to steganalysis, and automatically expresses the features of steganalysis effectively through learning. Different from traditional methods based on artificial design features, this method integrates feature extraction module and classification module into a trainable network model framework, and automatically learns features and realizes classification in the form of data-driven, thus greatly reducing the need for human experience and time. On this basis, the detection performance is further improved by model fusion and other methods.

2. A steganalysis feature learning method based on global statistical information constraints is proposed.

This work focuses on the importance of global information in steganalysis feature expression, and introduces how to use global statistical information to make the model learn better feature expression under the framework of feature-based learning steganalysis. Referring to the idea of transfer learning, this paper proposes a CNN model based on global statistical information constraints. By calculating auxiliary features to obtain additional global statistical information, and then introducing global statistical information into CNN model in the form of auxiliary tasks in migration learning, regularization constraints are applied to training CNN model, so that it can learn better feature expression.

3. A feature learning method for low embedding rate steganalysis is proposed.

In the field of steganalysis, the detection of low embedding rate encrypted images is a key concern. The difficulty lies in the small amount of embedding information, which makes the change of image statistical characteristics relatively small, so it is more difficult to detect. To solve this problem, based on the idea of Transfer Learning, this paper proposes to enhance the feature learning on

low embedding rate dense image datasets by migrating the prior information of features learned from high embedding rate dense image datasets of CNN, so as to improve the detection performance of CNN model for low embedding rate dense image.

4. A general steganalysis method for multi-class steganography is proposed.

From the current research status of general steganalysis algorithms, the “universality” of existing methods only reflects that different steganalysis algorithms can use the same algorithm steps to construct detectors. However, in the process of constructing the detector, we need to know the specific types of edge information of the steganography algorithm to be detected, and use these edge information to generate the dense image for training. There are many possible steganography algorithms in real application scenarios. Steganalysts often find it difficult to know which steganography users use. Therefore, it is urgent and significant to study steganalysis detectors for different steganalysis algorithms (known or even unknown). This chapter will explore this issue. On the basis of previous research on Steganalysis Based on in-depth learning and combined with the idea of multi-task learning, this paper tries to propose a steganalysis method for steganographic images generated by different steganographic algorithms.

2. Relevant research summary

Neil E. Johnson of George Mason University in the late 1990s in the United States first began the study of steganalysis. Later, Dartmouth College, Massachusetts Institute of Technology, New York State University, Purdue University, New Jersey Institute of Technology, Wet Stone Corporation, IBM Corporation, Microsoft Corporation and other institutions have carried out research in this direction, and most of them have received strong support from the United States Department of Defense, the National Security Agency and other government departments. In addition, the United Kingdom, Germany, France, Russia, Japan, Finland and other countries have also actively invested in the research and published important results. At present, Jessica Fridrich, Andrew D. Ker, Yun Q. Shi, Haw Farid, Niels Provos, Andreas Westfeld and other well-known research experts are in the field. In China, the Institute of Automation, the Institute of Information Engineering, the Chinese Academy of Sciences, the Beijing Institute of Electronic Technology Application, Tsinghua University, Sun Yat-sen University, China University of Science and Technology, Shenzhen University, Beijing University of Posts and Telecommunications, Dalian University of Technology, Tianjin University, Shanghai University, Tongji University, PLA University of Information Engineering, Hunan University and other universities and research institutes are all located. The research has been carried out and supported by national “863”, “973”, “National Natural Science Foundation” and other projects or funds. International well-known journals and conferences in the field include: IEEE Transactions on Information Forensics and Security (TIFS), Information Hiding and Multimedia Security (IH&MMSec), Media Watermarking, Security, and Forensics (MWSF), International Workshop on Information Forensics and Security (WIFS), International Workshop on Digital Watermarking (IWDW), etc. As of October 2016, the National Conference on Information Hiding and Multimedia Information Security has successfully held 13 sessions.

Although steganalysis research has made continuous progress in recent years, the so-called “one foot tao, one foot devil tail”, steganography research is also developing rapidly, and has brought new challenges to steganalysis. Advanced steganography can make

better use of the variability and complexity of natural image content to conceal the existence of weak steganographic signals. At the same time, the existence of large amounts of image data on the Internet helps to make the carrier images with hidden information relatively safe to be mingled among them. These factors make steganalysis like “looking for a needle in a haystack”. The increasing difficulty of steganalysis also makes the traditional heuristic-based Steganalysis Based on artificial design features face a more severe test. Therefore, there is an urgent need to develop some new methods for steganalysis. In recent years, with the rise of in-depth learning methods, feature learning has gradually been the focus of machine learning research. In-depth learning uses deep neural networks to automatically learn effective feature representation in the form of data-driven, and quickly overturns the research framework based on artificial features in many machine learning related application fields, becoming a new research paradigm. The idea of in-depth learning is expected to bring new research ideas for steganalysis tasks. As a confrontational technology, the research of steganalysis needs to pay attention to the development of steganalysis, so as to achieve “targeted”, so this chapter first introduces the research status of steganalysis, and then introduces the development of steganalysis. At the same time, in order to better understand the in-depth learning methods used in this paper.

3. Steganalysis based on convolutional neural network

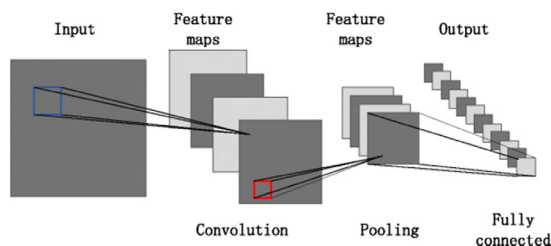
3.1. Introduction

At present, the most concerned problem in steganalysis is feature expression. Good feature expression plays a key role in the final detection performance. In traditional steganalysis methods, this part is done by human. However, it is a very laborious and heuristic method to select features manually. Whether effective features can be selected depends largely on human experience, and the adjustment of feature parameters takes a lot of time and energy. Up to now, although great progress has been made in feature extraction of steganalysis, many effective features have emerged. Because of the difficulty of steganalysis itself and the continuous development and progress of steganography, the design of steganalysis features has brought serious challenges. Especially in recent years, the content adaptive steganography algorithm can maintain more complex statistical characteristics such as neighborhood correlation, which makes steganalysts need to consider more complex image statistical characteristics for effective detection, and the difficulty of artificial design features is also increasing, which puts forward higher requirements for Designers' experience and knowledge.

In view of the problems faced by steganalysis, unlike traditional methods, this work pioneered the introduction of in-depth learning into steganalysis [20–23]. Considering the problem of feature expression in steganalysis from the perspective of feature learning, a new idea to solve the problem of steganalysis was proposed. In this chapter, a steganalysis framework based on in-depth learning is proposed. Based on the representative deep learning model, Convolutional Neural Network (CNN), combined with the characteristics of steganalysis task, the network structure is further designed and selected, which makes the deep learning model more suitable for learning the feature expression related to steganalysis. Particularly noteworthy is that this chapter proposes a new framework for steganalysis, which proves the validity of feature-based learning based on in-depth learning model in steganalysis through experimental analysis. It has guiding and referential significance for the next development direction of steganalysis.

3.2. Steganalysis feature learning framework based on CNN.

The feature of in-depth learning is that the complex representation is fitted by multi-layer non-linear transformation structural units, and the effective feature expression is learned automatically through training. For specific application areas, the key to the success of in-depth learning is whether the structure design of in-depth model is suitable for capturing task-related statistical characteristics. At present, the most concerned statistical feature in steganalysis research is neighborhood dependencies. In fact, with the continuous development of steganography, the new steganography algorithm can maintain some simple statistical characteristics of images, but it is difficult to completely maintain complex statistical characteristics such as neighborhood correlation. At present, the most effective artificial features in the field of image steganalysis are based on the modeling of neighborhood correlation to distinguish the changes brought about by steganographic embedding operations. Due to the lack of accurate image statistical models, and the current advanced content adaptive steganography usually embeds information into image regions with complex texture and noise, the neighborhood correlation of these regions is more complex, and it is difficult to effectively model. In traditional methods, feature design patterns based on human experience are facing great challenges. In this study, by using the successful experience of traditional steganalysis methods for reference, the important information of neighborhood correlation is also taken into account in the selection and structure design of deep learning model, so that the designed deep learning model can automatically mine the neighborhood correlation of steganalysis and learn effective feature expression. Therefore, a steganalysis model based on CNN is proposed. CNN, as a representative in-depth learning method, is a kind of neural network with special network structure. It contains local connection, weight sharing, downsampling and other structural design ideas. It can directly learn more effective feature expression from data. From the perspective of steganalysis, the convolution structure in CNN is conducive to capturing the high-order statistical information of the correlation between adjacent pixels, which is very important for steganalysis. At the same time, weight sharing, pooling and other structures also greatly reduce the training parameters, making the network model can handle larger size and number of pictures. Therefore, CNN is selected as the basic model in the framework of deep learning in this paper. Furthermore, on the basis of the traditional CNN structure, this work takes more into account the characteristics related to steganalysis tasks. By designing and selecting appropriate model structure units and parameters, the proposed model can effectively learn the expression of steganalysis features.



4. Steganalysis feature learning based on global statistical information constraints

4.1. Introduction

In the previous chapter, a CNN-based steganalysis framework is proposed, which can automatically acquire effective steganalysis

feature expression by feature learning. In this chapter, further research is carried out on the basis of previous research work. This work focuses on the importance of global information in steganalysis feature expression, and introduces how to use global statistical information to enable the model to learn better feature expression under the framework of feature-based learning steganalysis. Global statistical information in steganalysis generally refers to the statistical information based on the whole image, which plays an important role in steganalysis. Generally, steganography can be seen as adding noise to an image. At the same time, the noise is very weak relative to the image content itself, and can be easily masked by the complex image content itself. From the long-term research experience of steganalysis, steganalysts need to consider two aspects of information at the same time in order to better capture the slight changes brought by the modification of steganalysis operations in some local areas of the image. The first aspect is the local information, that is, how to describe the relationship between the elements in the local area of the image. It often involves image characteristics such as complex correlation between pixels. The second aspect is global information, that is, considering the relationship between local elements of the image, further global image statistics is needed to make the image changes caused by steganography more significant in the image feature space. Both of them play an extremely important role in steganalysis.

Through the analysis of the steganalysis model based on CNN proposed in the previous chapter, this paper considers that the convolution structure in the model has advantages in capturing the correlation between the pixels in the local area of the image, but in the use of global information in the process of feature extraction, CNN usually fuses the information of the local area layer by layer through the scaling of the convolution layer or the pooling operation, which exists in itself. Restrictions. To solve this problem, this chapter proposes a method based on model regularization constraints. The goal is to use model regularization technology to introduce global information and achieve more effective steganalysis feature learning.

4.2. Relevant work

Researchers have long noticed the importance of global image statistics for steganalysis. In traditional methods based on artificial features, how to use global information has always been a problem to be considered in feature design. In early steganalysis feature extraction, image global statistical histogram and statistical moments based on histogram are usually used as features. These methods make use of the global statistical characteristics of images to distinguish the differences between images before and after steganography.

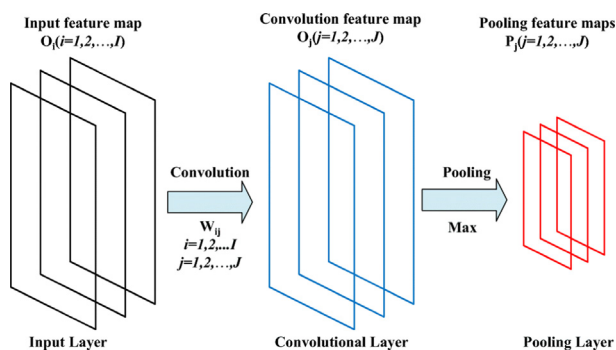
In the steganalysis method based on CNN proposed in the previous chapter, the convolution structure of CNN is suitable for describing the relationship between the pixels in the local area of an image, and through hierarchical convolution, it can automatically learn more complex image local relationship patterns. In terms of global information utilization, it is different from traditional steganalysis methods. In the CNN model, the local area information of the image is fused by pooling. Through multi-layer pooling operation, the high-level features of CNN fuse the local information in a larger range of images from the low-level layer by layer. Compared with the traditional global statistical methods such as histogram and co-occurrence matrix, pooling is a region-based statistical method. It is different from the fine-grained statistics of calculating multiple bin (groups) from the whole graph in histogram and co-occurrence matrix. In pooling, a response value is calculated from the corresponding region, and it is easy to lose some useful global information for steganalysis. This chapter tries

to study how to make better use of global information and promote more effective feature learning based on CNN model.

4.3. Steganalysis feature learning model based on global information constraints

In order to make better use of global statistical information in steganalysis feature learning and use the idea of transfer learning for reference, a CNN model based on global statistical information constraints is proposed in this paper. By calculating auxiliary features to obtain additional global statistical information, the information is introduced into the CNN model in the form of auxiliary tasks in transfer learning, and the training of CNN model is regularized to make it learn better feature expression.

The method proposed in this paper enlightens the successful application of self-transfer learning in some other research fields based on in-depth learning. Transfer learning is a machine learning method that helps new task learning by transferring knowledge in a task or data domain. The key to its success lies in the fact that the first two tasks are related and have some commonalities. At the same time, one task can provide information useful to another task and not easily captured by itself. In the field of target recognition, Oquab et al. proposed the task of transferring the mid-level image feature expression learned from supervisory training on large-scale data sets to target recognition on small-scale data sets. The method is to use some deep network layer parameters after training on large-scale data sets when training on small-scale data sets. Ji et al. used similar ideas in the research of video-based human behavior recognition. In this method, the behavior information of long continuous frames is introduced into the 3D-CNN model through auxiliary motion features. There are other similar tasks that utilize auxiliary tasks or features. Unlike these works, this work focuses on steganalysis. The goal of this work is to introduce global statistical information into CNN model to improve the performance of model detection by referring to the idea of transfer learning.



5. Feature learning for low embedding rate steganalysis

5.1. Introduction

In the previous two chapters, the most concerned feature expression problem in steganalysis is studied from the perspective of feature learning. A steganalysis model based on CB_r1N is proposed, and the expressive ability of the model is improved by means of model structure design and prior information introduction. On the basis of previous research work on steganalysis feature learning based on CNN, this chapter focuses on steganalysis feature learning for low embedding rate steganalysis image detection.

The difficulty of steganalysis is that the steganographic noise signal introduced into images by steganographic operations is usually very weak, and the difference between images before and after steganographic operations is very small, and this difference can easily be masked by the difference between different image contents. When the amount of information embedded in the image decreases gradually, the difficulty of steganalysis will be further increased. In fact, Steganalysis for the detection of low embedding rate encrypted images has always been a challenging problem. The difficulty is that, compared with high embedding rate embedding, low embedding rate embedding requires fewer image elements to be modified and fewer changes to the statistical characteristics that may be used by steganalysts. Especially in recent years, the content adaptive steganography proposed in this paper can hide steganographic noise signals into complex texture regions of images which are difficult to model statistically when embedding information. Therefore, under the condition of low embedding rate, adaptive steganography can give full play to its advantages and is more difficult to detect.

Based on the research work of steganalysis feature learning in the previous chapters, this paper finds that the training difficulty of the proposed steganalysis model based on CNN is especially obvious when faced with low embedding rate steganalysis task, and even cannot converge. It is worth emphasizing that the standard of low embedding rate is not accurately defined in steganalysis. The main reason is that the difficulty of steganalysis needs to consider both the steganographic algorithm to be detected and the embedding rate, and it is difficult to define the absolute low embedding rate according to the detection difficulty. The low embedding rate mentioned in this paper is a relative concept and relative to the same steganography algorithm. Although the detection difficulty may vary greatly under the condition of uniform embedding rate of different algorithms, for the same algorithm, the detection difficulty will gradually increase with the decrease of embedding rate until it is difficult to distinguish between normal image and dense image. Experiments show that the training of CNN model cannot converge when the embedding rate of the classified image in the training set is less than 0.1 BPP when training and detecting steganography with the CNN model proposed in Chapter 3. In fact, although deep learning networks such as CNN have good feature learning ability, there are a lot of training parameters in these deep learning network models, which are easy to fall into local optimum in the training process, so there are some training difficulties. This situation is particularly evident in the training of small data sets. For steganalysis tasks, the difference between the encrypted image and the normal image is smaller under the condition of low embedding rate. At the same time, the content of different images is greatly different, which makes CNN more easily misled by the difference of image content, and it is difficult to distinguish steganographic noise signals. Aiming at this problem, based on the idea of transfer learning, this work proposes to enhance the feature learning on low embedding rate dense image data set by transferring the prior information of the features learned from high embedding rate dense image data set of CNN, so as to improve the detection performance of CNN model for low embedding rate dense image. This method has a good effect in the face of the most advanced content adaptive steganography.

5.2. Analysis of content adaptive steganography

In order to better elaborate the research methods proposed in this chapter, the characteristics of content adaptive steganography algorithm are introduced and analyzed before the research methods are specifically introduced. Content adaptive steganography is one of the most secure steganography algorithms. The core idea

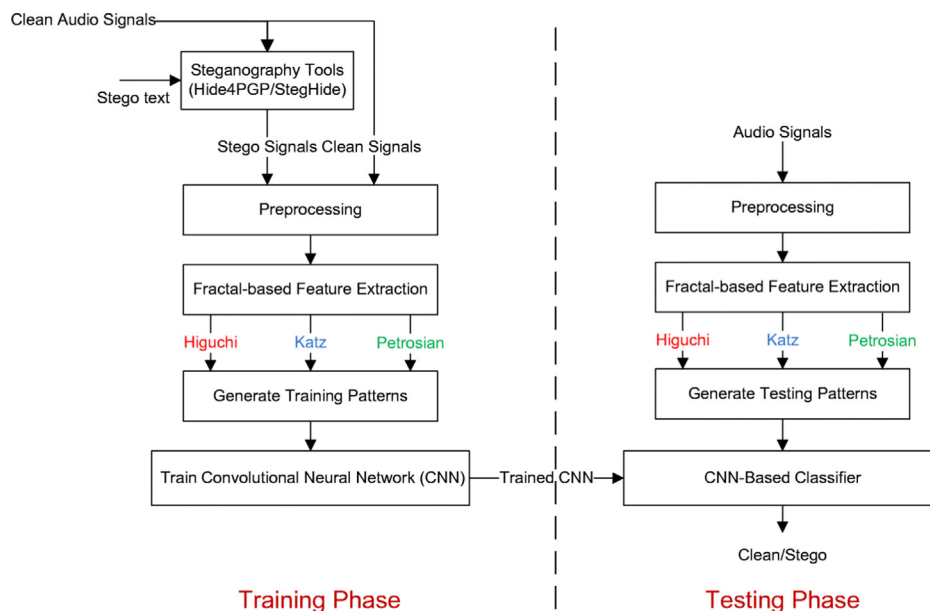
of this kind of algorithm is to make use of the complex statistical characteristics of texture regions in images, which make it difficult to model effectively. For a given embedding rate, the information is embedded in texture regions first, so that the influence of steganography modification can be covered up by the complex statistical characteristics of texture.

Fig. 5.1 shows the embedding position of information after embedding a carrier image in a representative content adaptive algorithm database according to different embedding rates. The carrier image contains the rough surface of the cliff, the vegetation layer on the cliff, the edge of the building and other areas with complex texture, as well as the smooth areas of the sky and the building wall. When embedding information with WOW, it can be found that when embedding rate is low, the information is basically embedded in the region with the most complex texture orientation. With the increase of embedding rate, the location of embedding information gradually expands from texture-rich regions to relatively smooth regions. For example, when the embedding rate is 0.6 bpp, part of the information is embedded in the smoother wall area between the building and the cliff. The map reflects the priority of embedded region selection in content adaptive algorithm, that is, according to the texture complexity from high to low. This kind of steganographic algorithm has more advantages in low embedding rate, and it is more difficult to detect than the early steganographic algorithm which randomly chooses the embedding area.

Another feature of this kind of algorithm is that, in the process of increasing embedding rate, the location of information embedding usually diffuses gradually along the extension of the region with the richest texture. That is to say, the embedding area corresponding to high embedding rate generally includes the embedding area corresponding to low embedding rate, and the embedding area with relatively low embedding rate can be regarded as the extension of texture area used in low embedding rate. From the perspective of steganalysis, since high-embedding usually first uses texture regions similar to low-embedding, the statistical characteristics of images affected by high-embedding are similar to that of low-embedding. At the same time, higher embedding modifies more image elements, leaving more “traces” that can be used for steganalysis.

5.3. Method

Through the research and analysis of the characteristics of content adaptive steganography in the previous section, we find out the reasons why it is difficult to detect content adaptive steganography in low embedding rate. At the same time, we find that there are some commonalities in the influence of different embedding rates on the statistical characteristics of images. This paper considers that there is a priori information that can be used by low embedding rate steganalysis in high embedding rate steganographic images which are relatively easy to detect. Based on the idea of transfer learning, this paper proposes to transfer auxiliary information from high embedding rate image to CNN model training for detecting low embedding rate image. This method is different from the traditional steganalysis, which treats different embedded images independently. In traditional steganalysis, when steganographic images with different embedding rates are detected by the same steganographic algorithm, independent training classifiers are usually trained according to different embedding rates. The proposed method hopes to introduce information from high embedding rate images to help low embedding rate dense image detection. In the previous chapter, the transfer learning method is also used to introduce the priori information of auxiliary tasks into the CNN model. It uses auxiliary features and achieves the constraints of prior information by means of mixed training of auxiliary tasks and target tasks. This transfer learning method of mixed training of auxiliary task and target task has similar application in some computer vision tasks. Unlike the methods in the previous chapter, the goal of this chapter is to improve the detection performance of low embedding rate steganography, and the way of transferring information is similar to pre-training without using additional auxiliary features. Pre-training ideas similar to those in this paper are used in some computer vision work such as target recognition. The proposed framework consists of two modules, which correspond to target task and source task respectively. The target task is low embedding rate image detection, and the source task is also called auxiliary task, which is high embedding rate image detection. Firstly, the source task is trained, then the trained features are transferred to the target task, and the target task is trained to improve the training performance of the target task.



6. General steganalysis for multi-class steganography

6.1. Introduction

The ultimate goal of general steganalysis is to construct a steganalysis detector without distinguishing specific types of steganalysis algorithms, which can simultaneously detect different types of known or even unknown steganalysis. However, from the current research status of generic steganalysis algorithms, the “universality” of existing methods only reflects that different steganalysis algorithms can use the same algorithm steps to construct detectors. However, in the process of constructing the detector, we need to know the specific types of edge information of the steganography algorithm to be detected, and use these edge information to generate the dense image for training. In fact, the detection performance of steganalysis algorithms given in most previous studies is the result of “one-to-one” scenarios, and then the steganalysis image embedded with the same steganalysis algorithm in the training set is detected with the detector of Training I. At present, the general steganalysis research mainly focuses on this “one-to-one” framework for the following reasons. Firstly, the image modification caused by steganography is usually very weak, which is easily concealed by the difference between the content of the image itself. This also makes it difficult to establish an effective image statistical model in steganalysis. Researchers hope to use more edge information to build a more accurate image statistical model. Secondly, the influence of different steganography embedding mechanisms on the statistical characteristics of images is also very different. Similar to the problem caused by image heterogeneity in image classification, when constructing detectors in steganalysis, a similar “heterogeneity” problem needs to be considered if multiple steganographic algorithms are considered at the same time. In fact, due to the deficiencies of feature generalization performance in traditional feature-based steganalysis methods, there are great challenges to solve these problems. However, from the practical point of view, the “one-to-one” steganalysis model is not enough. There are many possible steganography algorithms in real application scenarios. Steganalysts often find it difficult to know which steganography users use. Therefore, it is urgent and significant to study steganalysis detectors for different steganalysis algorithms (known or even unknown). In this chapter, we will explore this issue. On the basis of previous research on Steganalysis Based on in-depth learning and combined with the idea of multi-task learning, a steganalysis method for steganographic images generated by different steganographic algorithms is proposed.

6.2. Relevant work

At present, in the field of image steganalysis, there are relatively few studies on the construction of detectors that can simultaneously detect different steganographic algorithms, especially for the early JPEG domain steganographic algorithms. From these works, we can summarize three kinds of solutions to problems. In the first method, the training data set is composed of cover image and stego image, which are generated by various steganography algorithms. Then the feature is extracted from the training data set and a two-class classifier is trained to distinguish the normal image from the stego image. In Pevny's and Fridrich's work, four different JPEG domain steganography algorithms are selected for hybrid training. After extracting DCT domain features, two classes of SVM classifiers are trained. So the trained classifier can distinguish the normal image from the image generated by different steganographic algorithms, even for the steganographic algorithm which is not used in training, it has certain detection

effect. This method is a more intuitive mixed training mode. In the second method, the training data set is constructed in the same way as in the first method, and the steganalysis features are extracted. The difference is that this method regards steganalysis as a multi-class task, and trains multi-class classifiers to distinguish between normal images and specific embedding algorithm types. In the method proposed by Pevny and Fridrich, seven JPEG domain steganography algorithms and dense images are generated according to different embedding rates. The training set is composed of normal images, and representative DCT domain features are extracted for training. In multi-classification tasks, the multi-classifiers constructed by them are used to distinguish seven categories including normal image classes and six types of steganography algorithms. In the training of multi-classifier, they adopt the method of “max.wins”. That is to say, they train 21 s-class SVMs for seven classes of images, and count the results of 21 s-class classifications. The classes that get the most votes in seven classes are the final decision types of multi-classifications. The main objective of this kind of method is to distinguish steganographic algorithm types concretely. The disadvantage of this method is that it needs to train several binary classifiers to distinguish two sets of binary classifiers. When the number of steganographic types increases, the computational cost of this method is high. The third method is to use one. class Classification. In this method, the training data set contains only the normal image, but does not use the encrypted image. Then a class of classifier is trained with the features extracted from the normal image, and the boundary of the normal image class in the feature space is found by training.

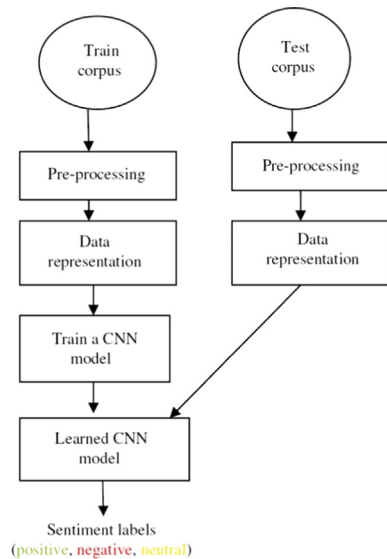
SVM detects steganography in JPEG domain. In addition to OC-SVM, Pevny and Fridrich tried One Class Neighbor Machine (OC-NM). Compared with the other two methods, this type of method can be trained independently of steganographic algorithm, but the problem is that the decision boundary of one class of classification is more difficult to determine than that of two and multi-class classification, so the detection effect of these methods is not ideal. In addition, the existing content adaptive steganography and other advanced methods can maintain the statistical characteristics of complex natural images, which will bring greater challenges to this type of methods.

By analyzing the advantages and disadvantages of various methods, the goal of this chapter is to solve the problem of binary classification. Even if the detectors that have to be constructed can successfully distinguish between cryptographic carriers and normal carriers in the face of images embedded in different steganography algorithms. Among the three methods mentioned above, the performance of multi-steganographic detector has an important relationship with feature expression, but these methods all depend on the features extracted manually. In the first and second methods, the steganalysis features corresponding to different steganographic algorithms are extracted separately, and the related characteristics between different algorithms are not utilized in the process. With the help of steganalysis research based on in-depth learning, this work incorporates feature expression, an important module, into the learning process. At the same time, drawing on the idea of multi-task learning, this paper hopes to use the complementary information between different steganography algorithms to improve the detection performance and generalization ability of detectors in the process of feature learning and classification.

6.3. Steganalysis framework based on multitask CNN

This section introduces the proposed steganalysis framework based on Multitask CNN. Firstly, the training set of multi-steganographic algorithm is constructed, and then the CNN model is trained by multi-task learning to optimize the binary classifica-

tion task of normal image/secret image and the classification task of multi-steganographic algorithm. Steganographic classification task is an auxiliary task. Its purpose is to help the main task, i.e. image/dense image binary classification task training. The final trained detector can detect many steganographic algorithms at the same time. How to choose the structure of CNN model and how to carry out multi-task learning in the whole framework is related to the final detection performance of the method.



7. Summary

With the rapid development and popularization of computer and Internet technology, the problem of information security on the network has become increasingly prominent. Steganography, as a representative covert communication technology, can be used illegally by terrorist organizations, spies and criminal gangs while ensuring communication security, thus endangering national security, public and personal privacy security and other issues. Therefore, the research of steganalysis technology for detecting and blocking steganography is of great significance.

Although steganalysis has developed rapidly in recent years, it still faces many difficulties and challenges. Based on the theory of in-depth learning method and image-based general steganalysis, this paper makes a deep study of the hot and difficult problem of steganalysis feature expression, and tries to establish a new steganalysis paradigm from the idea of feature learning. The main contributions of this paper are as follows:

1. An innovative steganalysis paradigm based on in-depth learning is proposed. In the field of steganalysis, feature expression is the focus and difficulty. Traditional methods depend on artificial design features. With the challenge of steganography, artificial design features become more and more difficult. This paper introduces in-depth learning into steganalysis from the perspective of feature learning. Based on the representative deep learning method CNN, the model is designed and adjusted according to the characteristics of steganalysis, which makes the proposed model more effective in capturing the statistical characteristics such as neighborhood correlation Related to steganalysis, and automatically learns the effective expression of steganalysis features, thus greatly reducing the need for human experience and time. In order to better understand the working principle of CNN for steganalysis, this paper visualizes the features learned, and

finds that the high-pass-like filter core is more conducive to capture the patterns related to steganographic noise. At the same time, in order to improve the generalization ability and detection performance of the model, this paper considers the method of model fusion, trains different CNN models separately, and fuses the output of the model. The experimental results show the effectiveness of the proposed method. The detection performance of the proposed method based on CNN is close to that of the traditional feature-based method. Finally, this paper analyses the performance of CNN using Ensemble Classifier to classify the features learned, and gets better detection performance by combining CNN features and artificial features.

2. A steganalysis feature learning method based on global information constraints is proposed. Based on the previous research of steganalysis method based on CNN, this work focuses on the importance of global information in steganalysis feature expression. By summing up the experience of traditional steganalysis methods, we have noticed that global statistical information plays an important role in steganalysis feature expression. In CNN model, the local region information of image is fused by pooling to make use of the information in a larger range, which is a region-based statistics. There are some limitations in using global statistical information. In order to make better use of global statistical information in steganalysis feature learning, this paper proposes to obtain additional global statistical information by calculating auxiliary features, then introduce global information into CNN model in the form of auxiliary tasks, and regularize the training of CNN model, so as to promote it to learn better feature expression. In the experiment, representative adaptive steganography WOW and S-UNIWARD are detected on BOSSbase library to verify the effectiveness of the proposed method. The results show that the detection performance of the proposed method based on global constraints is much better than that of the previous CNN-based method.

3. A feature learning method for low embedding rate steganalysis is proposed. Steganalysis for low embedding rate image detection has always been a challenging problem. Generally, the difference between the images before and after steganography is very small, and this difference is easily concealed by the difference between different image content, so it is difficult to detect. As the embedding rate decreases, the number of image elements that need to be modified in low embedding rate embedding is less, and the change of statistical characteristics that may be used by steganalysts is less, so it becomes more difficult to detect. Based on the research work of steganalysis feature learning in the previous chapters, experiments show that the proposed steganalysis model based on CNN is especially difficult to train and even unable to converge in the face of low embedding rate steganalysis task. To solve this problem, this work is based on the idea of transfer learning. First, the network is pre-trained by high embedding rate data sets, and then the features learned are transferred to low embedding rate detection model for further fine-tuning training. Through this method, the prior information of high embedding rate image is transferred to low embedding rate image detection, which improves the detection performance of CNN model for low embedding rate image. In the experiment, the effectiveness of the proposed method is verified by detecting WOW and UNIWARD steganography algorithms. The experimental results on BOSSbase database show that the method based on migration learning significantly improves the detection ability of steganalysis method based on CNN at low embedding rate.

4. A general steganalysis method for multi-class steganography is proposed. The ultimate goal of general steganalysis is to construct steganalysis detectors without distinguishing specific types of steganalysis algorithms. However, in the current steganalysis research, most of the methods cannot be truly universal. In the

process of constructing the detector, we need to know the specific type of edge information of the steganography algorithm to be detected, and use these edge information to generate the dense image for training. From the practical application point of view, the current method cannot meet the requirements. Therefore, it is urgent and significant to study steganalysis detectors for different steganalysis algorithms (even unknown ones). In this work, the problem is explored and studied. On the basis of previous research work on Steganalysis Based on in-depth learning, combined with the idea of multi-task learning, this paper attempts to propose a more general method for multi-steganalysis detection. This method constructs two related steganalysis tasks in CNN training, that is, the binary classification task to distinguish the normal image from the carrier image and the multi-classification task to distinguish the type of steganalysis algorithm. The goal of this work is to hope that “fine-grained” multi-classification tasks can introduce more abundant shared information into the model, and promote the performance of the two-category directors. In the experiment, four representative adaptive steganography algorithms are selected to generate a hybrid training set, and multi-task learning is used to train the known and unknown algorithms in the training set. The results show that the proposed method based on Multi-task has better performance than the direct hybrid training method.

Declarations

Ethical Approval and Consent to participate: Approved.

Consent for publication: Approved.

Availability of supporting data: We can provide the data.

Competing interests

These no potential competing interests in our paper. And all authors have seen the manuscript and approved to submit to your journal. We confirm that the content of the manuscript has not been published or submitted for publication elsewhere.

Author's contributions

All authors take part in the discussion of the work described in this paper. The author Ying Zou wrote the first version of the paper, and did part experiments of the paper, Ge Zhang initiated the project. Leian Liu revised the paper in different version of the paper.

Conflict of interest

There is no conflict of interest.

Funding

This work was supported by the Science and Technology Planning Project of Guangdong Province under Grant (2017A070709012), the quality resource sharing course project-“Computer Network” (Official document by Department of education of Guangdong province ([2015] no. 133)), the provincial-level characteristic specialty-“Network Engineering” and the provincial teaching team-“Teaching team of basic core course of computer major” (Official document by Department of education of Guangdong province ([2017]214)), 2018 higher education teaching reform project of Guangdong province-“Reform and practice of the training mode of network engineering talents based on the cooperation of school-school and school-enterprise” (Official document by Department of education of Guangdong province ([2018]

no. 180)), and the university-level high-quality resource sharing course project-“Network Security Technology” and “Wireless Sensor Network and RFID Technology”. And NSFC (No. 61802114, 61802113), Scientific Research Foundation of the Higher Education Institutions of Henan Province (18A520021, 18A120001), Project supported by the Jiangsu Key Laboratory of Image and Video Understanding for Social Safety (Nanjing University of Science and Technology No. 30916014107).

Acknowledgements

The authors thank the editor and anonymous reviewers for their helpful comments and valuable suggestions.

References

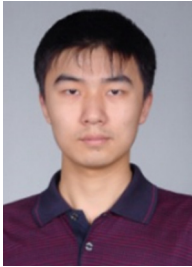
- [1] Neil F. Johnson, Sushil Jajodia, Exploring steganography: seeing the unseen., *Computer* 31 (1998) 2.
- [2] Jessica Fridrich, Miroslav Goljan, Du. Rui, Detecting LSB steganography in color, and gray-scale images, *IEEE Multimedia* 8 (4) (2001) 22–28.
- [3] Abbas Cheddad et al., Digital image steganography: survey and analysis of current methods, *Signal Process.* 90 (3) (2010) 727–752.
- [4] L. Zhang, M. Song, Z. Liu, X. Liu, J. Bu, C. Chen, Probabilistic graphlet cut: exploiting spatial structure cue for weakly supervised image segmentation., in: *Proceedings of the IEEE conference on computer vision and pattern recognition*, IEEE, Portland, 2013, pp. 1908–1915.
- [5] Cachin, Christian, An information-theoretic model for steganography, *International Workshop on Information Hiding*, Springer, Berlin, Heidelberg, 1998.
- [6] Lisa M. Marvel, Charles G. Boncellet, Charles T. Retter, Spread spectrum image steganography, *IEEE Trans. Image Process.* 8 (8) (1999) 1075–1083.
- [7] Donovan Artz, Digital steganography: hiding data within data, *IEEE Int. Comput.* 5 (3) (2001) 75–80.
- [8] Junwei Han, Dingwen Zhang, Gong Cheng, Lei Guo, Jinchang Ren, Object detection in optical remote sensing images based on weakly supervised learning and high-level feature learning, *IEEE Trans. Geosci. Remote Sens.* 53 (6) (2015) 3325–3337.
- [9] Weiqi Luo, Fangjun Huang, Jiwei Huang, Edge adaptive image steganography based on LSB matching revisited, *IEEE Trans. Inform. Forensics Secur.* 5 (2) (2010) 201–214.
- [10] L. Zhang, R. Hong, Y. Gao, R. Ji, Q. Dai, X. Li, Image categorization by learning a propagated graphlet path, *IEEE T-NNLS* 27 (3) (2016) 674–685.
- [11] Chang-Chou Lin, Wen-Hsiang Tsai, Secret image sharing with steganography and authentication, *J. Syst. Software* 73 (3) (2004) 405–414.
- [12] Hussain, Mehdi, Mureed Hussain, A survey of image steganography techniques, 2013.
- [13] Junwei Han, Xiang Ji, Hu. Xintao, Dajiang Zhu, Kaiming Li, Xi Jiang, Guangbin Cui, Lei Guo, Tianming Liu, Representing and retrieving video shots in human-centric brain imaging space, *IEEE Trans. Image Process.* 22 (7) (2013) 2723–2736.
- [14] Tuo Zhang, Lei Guo, Kaiming Li, Changfeng Jing, Yan Yin, Dajiang Zhu, Guangbin Cui, Lingjiang Li, Tianming Liu, Predicting functional cortical ROIs via DTI-derived fiber shape models, *Cerebral Cortex* 22 (4) (2012) 854–864.
- [15] L. Zhang, Y. Gao, Y. Xia, Q. Dai, X. Li, A fine-grained image categorization system by cellet-encoded spatial pyramid modeling, *IEEE Trans. Ind. Electron.* 62 (1) (2015) 564–571.
- [16] Dingwen Zhang, Deyu Meng, Junwei Han, Co-saliency detection via a self-paced multiple-instance learning framework, *IEEE Trans. Patt. Anal. Mach. Intell.* 39 (5) (2017) 865–878.
- [17] L. Zhang, Y. Gao, Y. Xia, K. Lu, J. Shen, R. Ji, Representative discovery of structure cues for weakly-supervised image segmentation, *IEEE Trans. Multimedia* 16 (2) (2014) 470–479.
- [18] Dingwen Zhang, Junwei Han, Chao Li, Jingdong Wang, Xuelong Li, Detection of co-salient objects by looking deep and wide, *Int. J. Comput. Vis.* 120 (2) (2016) 215–232.
- [19] Gong Cheng, Peicheng Zhou, Junwei Han, Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images, *IEEE Trans. Geosci. Remote Sens.* 54 (12) (2016) 7405–7415.
- [20] Sorina Dumitrescu, Xiaolin Wu, Zhe Wang, Detection of LSB steganography via sample pair analysis, *International Workshop on Information Hiding*, Springer, Berlin, Heidelberg, 2002.
- [21] Junwei Han, Dingwen Zhang, Hu. Xintao, Lei Guo, Jinchang Ren, Feng Wu Background prior-based salient object detection via deep reconstruction residual, *IEEE Trans. Circ. Syst. Video Technol.* 25 (8) (2015) 1309–1321.
- [22] Junwei Han, King Ngi Ngan, Mingjing Li, Hong-Jiang Zhang, Unsupervised extraction of visual attention objects in color images, *IEEE Trans. Circ. Syst. Video Technol.* 16 (1) (2006) 141–145.
- [23] L. Zhang, Y. Xia, R. Ji, X. Li, Spatial-aware object-level saliency prediction by learning graphlet hierarchies, *IEEE Trans. Indus. Electron.* 62 (2) (2015) 1301–1308.



Ying Zou was born in Henan, China in 1979. She is currently a lecturer in Zhongkai University of Agriculture and Engineering, China. She is a member of China Computer Federation. Her research interests include: network security technology, IoT technology and machine learning. In recent five years, as the first author has published 1 paper, chaired three software copyrights.



Leian Liu is an associate professor in computer science and technology, Zhongkai University of Agriculture and Engineering, China. He received the Ph.D. in Circuits and Systems from South China University of Technology. He is a member of Guangdong Institute of Electronics and China Computer Federation. Currently, he is interested in the research areas such as IoT technology, network security, and embedded technology, etc.



Ge Zhang received the B.S. and M.S. degrees in computer science from the Zhongnan University of Economics and Law, Wuhan, China, in 2005 and 2007, respectively, and received the Ph.D. degree from Huazhong University of Science and Technology, Wuhan, China, in 2015. Now he is an assistant professor in School of Computer and Information Engineering, Henan University, Kaifeng, China. His research interests include image processing, streaming media, network modeling, and measurement.